# DIPLOMA PROJECT PRESENTATION

## FEASIBILITY STUDY OF AN MLAT SURVEILLANCE SYSTEM

Supervisor: PhD. Eng. Cristian-Emil CONSTANTINESCU

Author: Stefan SUBASU

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# AGENDA

1) Introduction to Radar Theory
2) Multilateration Overview and Methods
3) Case study: ADS-B signal coverage in Romania
4) Location choosing
5) DEM of Romania
6) Theoretical and Real ADS-B signal range
7) Error study
8) Result interpretation
9) Conclusions and Future Research

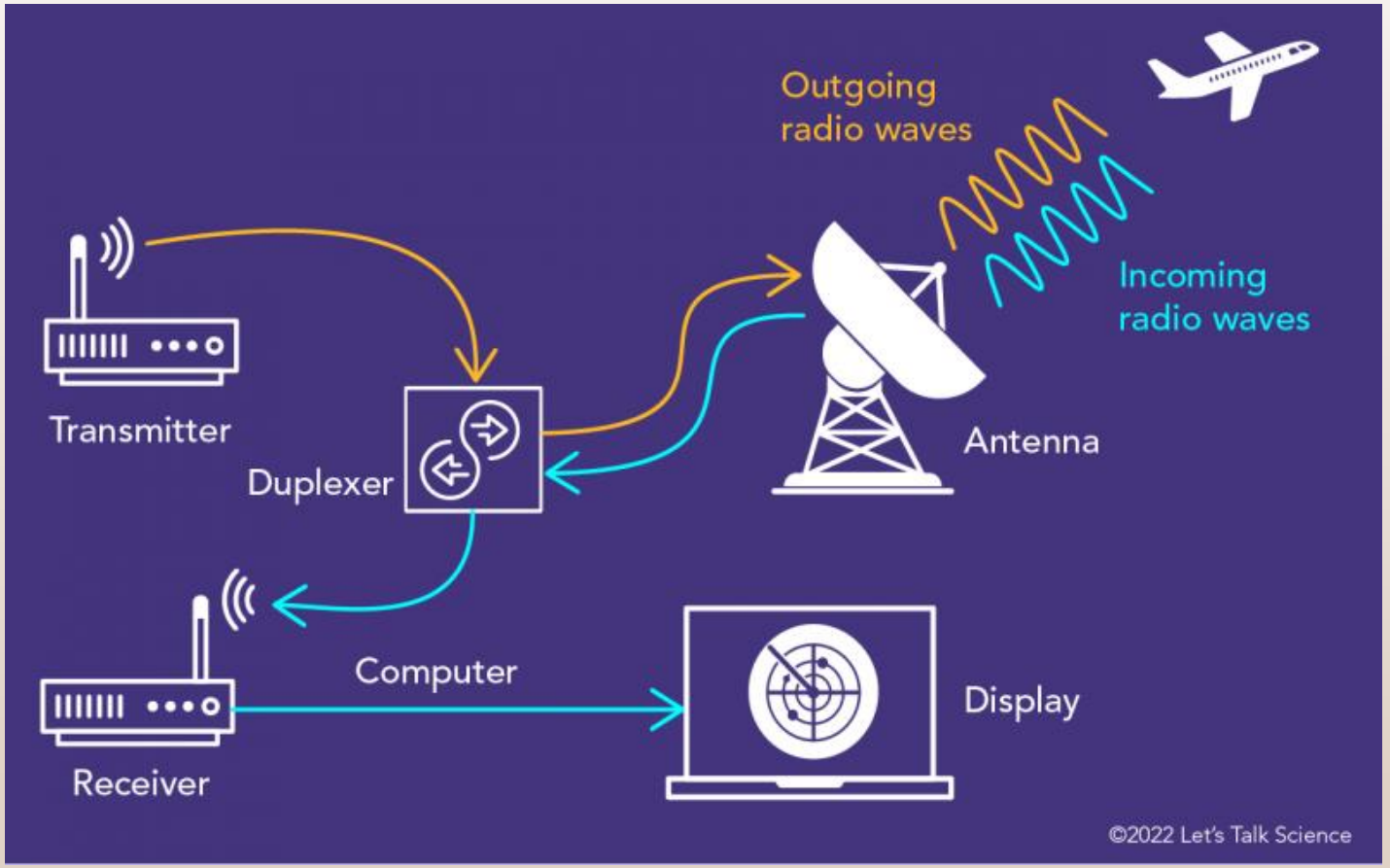
# Basic RADAR Theory

RADAR = RAdio Detection

And Range

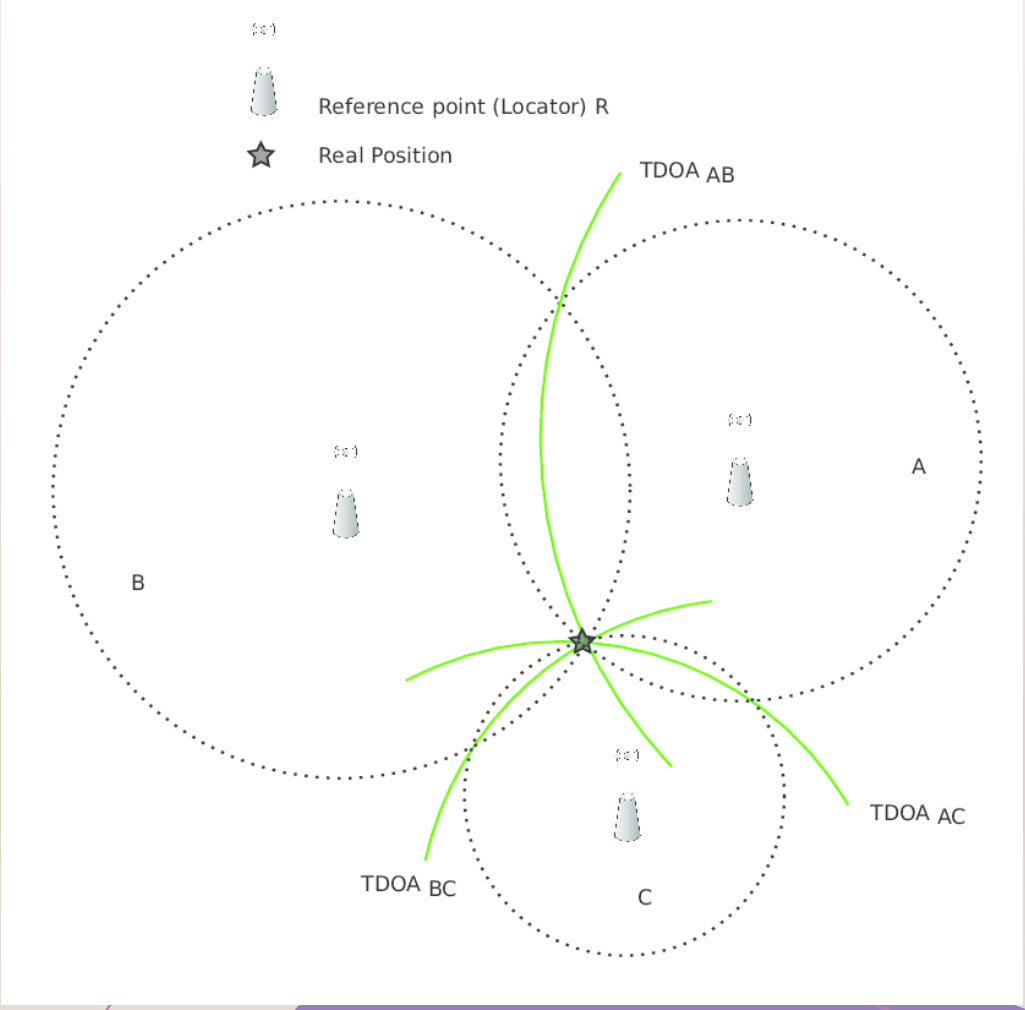
RADAR Equation

$$D_i = \frac{P_t g_t}{4\pi R^2}$$



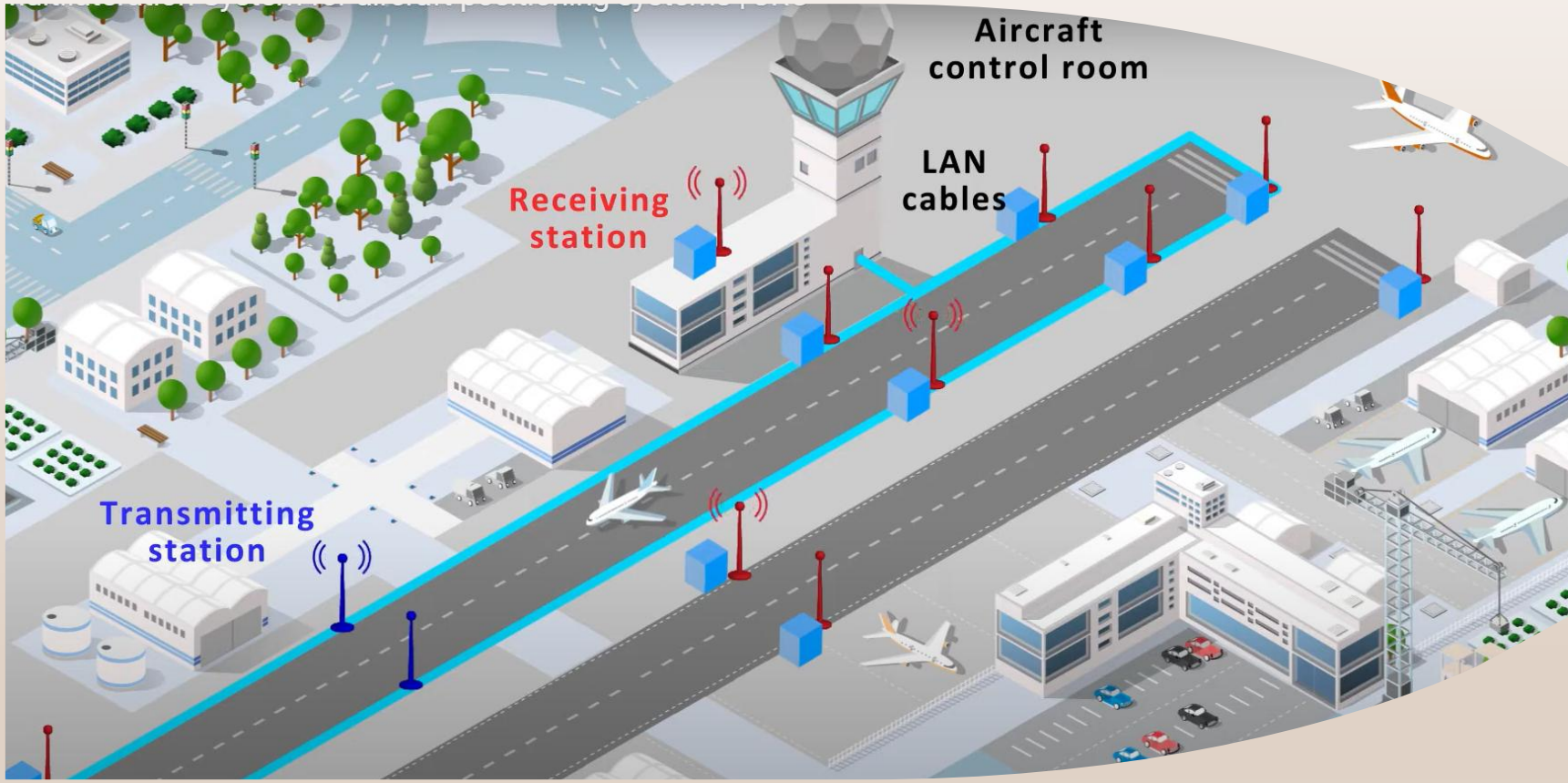
# Multilateration: Overview and Methods

- Hyperbolic multilateration
- Circular multilateration
- Spherical multilateration



# MLAT Surveillance System

- Signal Transmission
- Signal Reception
- Time Difference Calculation
- Aircraft Position Calculation



# Why ADS-B?

1) Higher precision and update rate
2) Superior coverage
3) Transmit more aircraft information
4) Increase capacity
5) Cost benefits

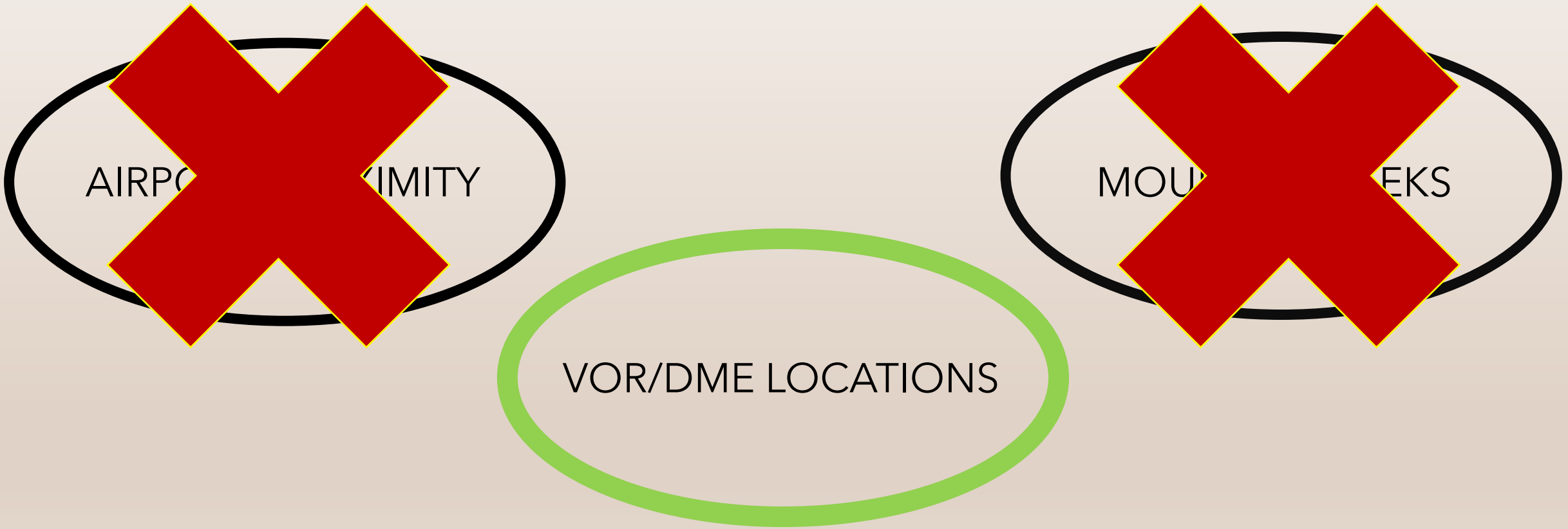


Thales AX680 antenna with triple redundancy

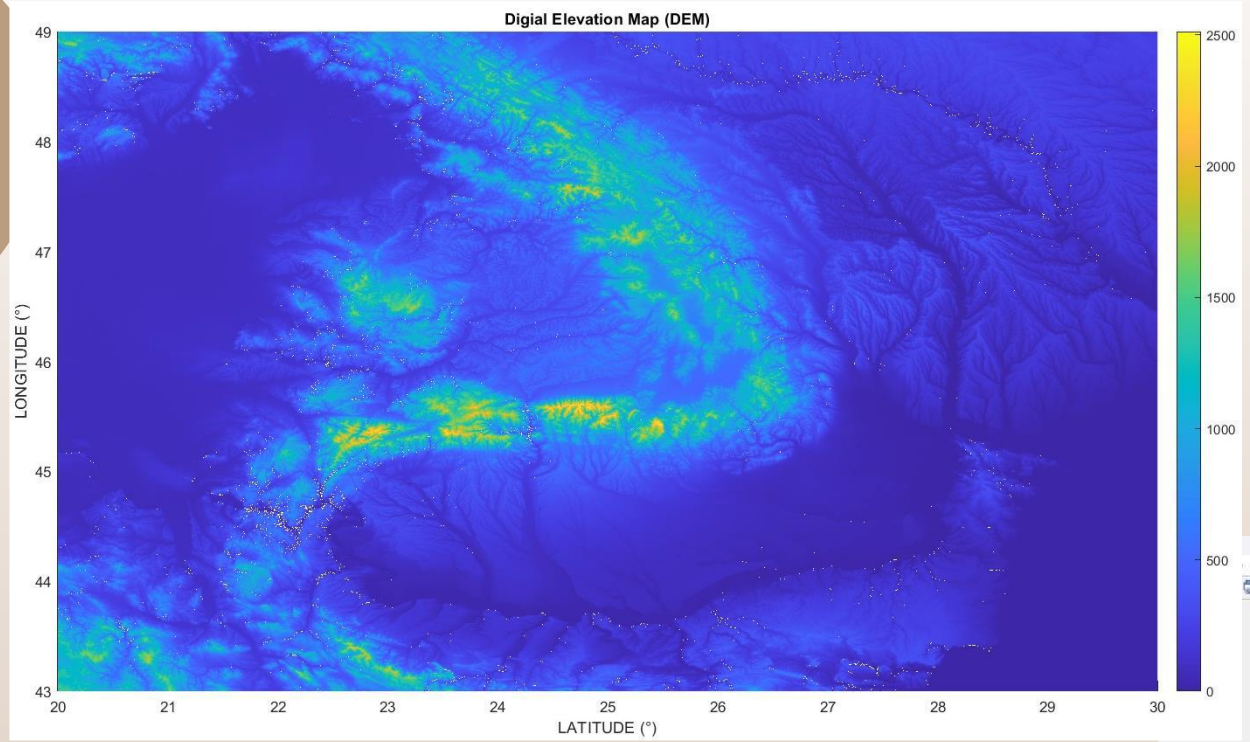
# ADS-B Antenna Coverage: Study Introduction

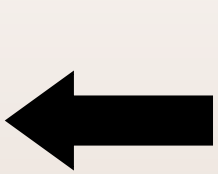
| What is the roadmap of the study? | What is the significance of the study? |
| --- | --- |
| Location selection for ADS-B antennas | Improving Aviation Safety |
| Consideration of Digital Elevation Model (DEM) | Increasing Efficiency of Air Traffic Control |
| Perform calculation of coverage for low altitude flight | Guiding Future Policy and Investment |

# What is the best location for ADS-B antennas?

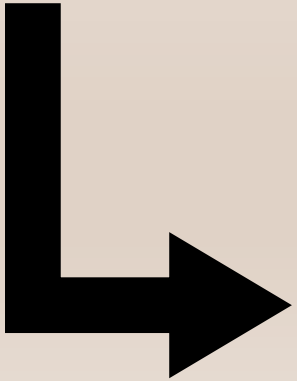


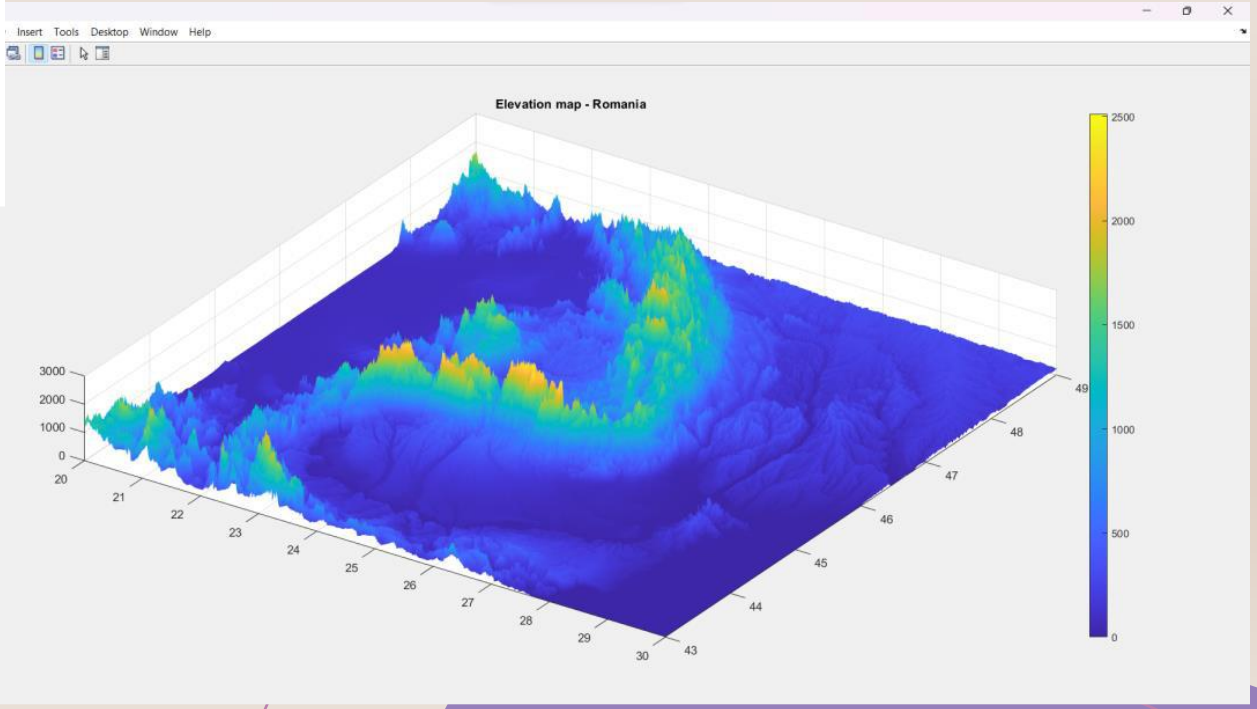
# DEM of Romania








**MATLAB code is referenced in APPENDIX A-1: MATLAB Simulation Code - Elevation Map**

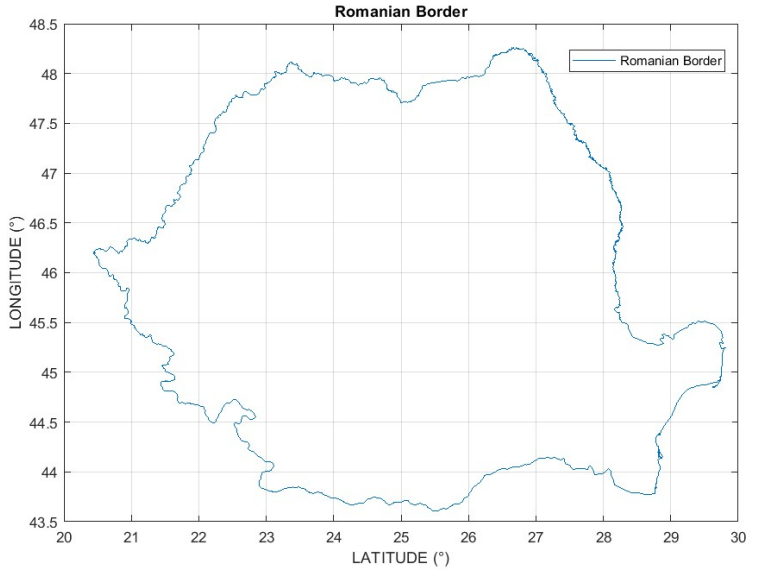
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# Romanian Border over DEM

```
% Load the country border shapefile
shapefile = 'C:\Users\stefa\Desktop\licenta poli\Shapefile\3\ro_frontiera_poligon.shp';
S = shaperead(shapefile);

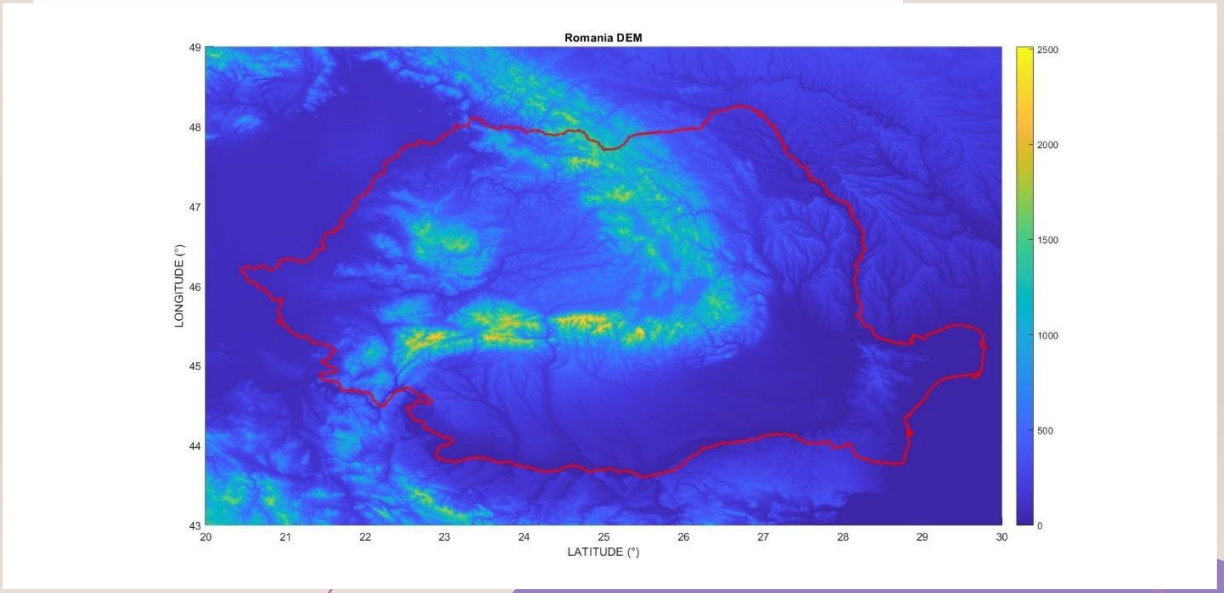
% Plot the border
latref=48.2460;
longref=26.7251;
lat=(S.Y-S.Y(1))/1852/60+latref;
lon=(S.X-S.X(1))/1852/60/cosd(45)+longref;

figure
plot(lon, lat)
title('Romanian Border');
legend('Romanian Border');
xlabel('LATITUDE (°)');
ylabel('LONGITUDE (°)');
grid on;
```

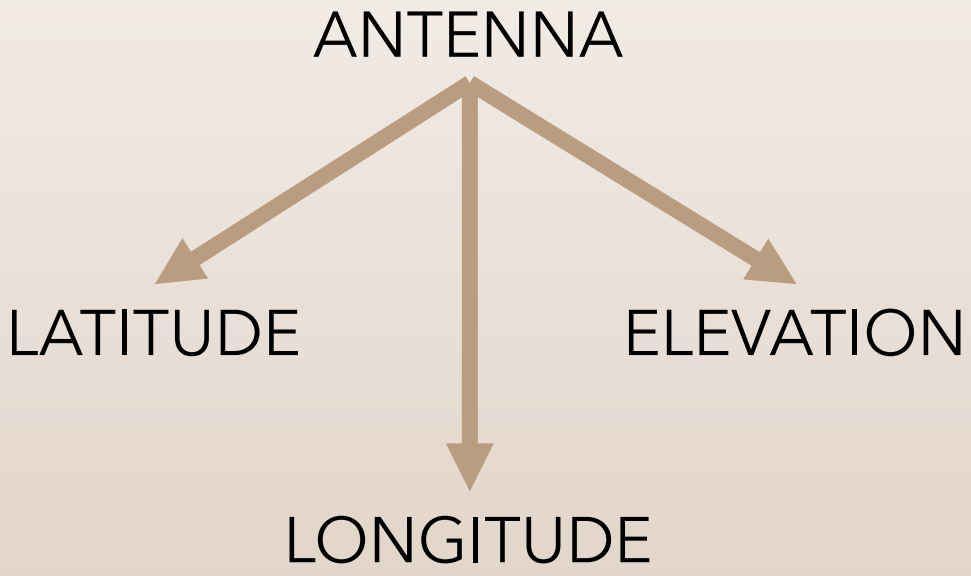



**MATLAB code is referenced in APPENDIX A-2: MATLAB Simulation Code – Romanian Border**

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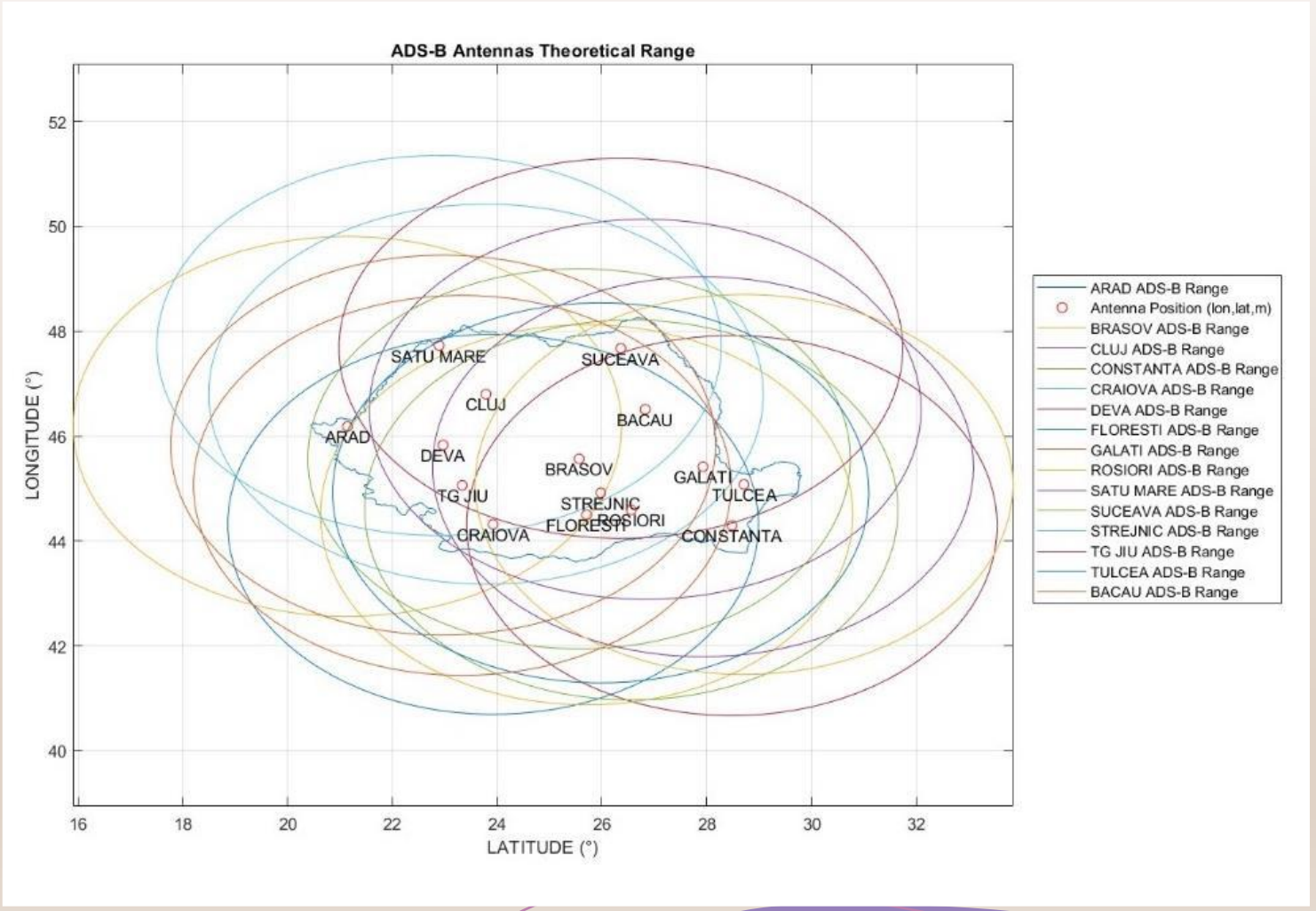


# ADS-B Theoretical Range




$$1\,\text{deg lon} \simeq 111\,\text{km}$$

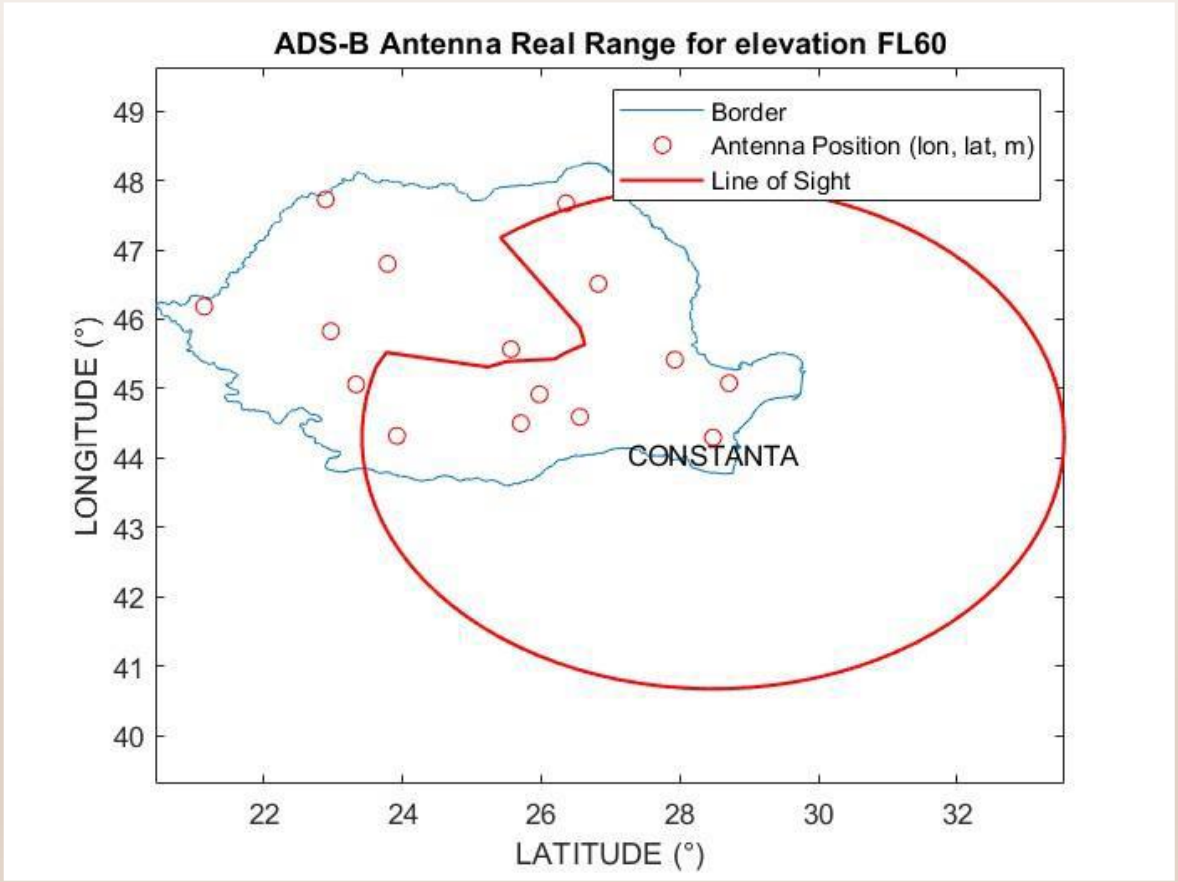
$$1\,\text{deg lat} \approx 78\,\text{km}$$

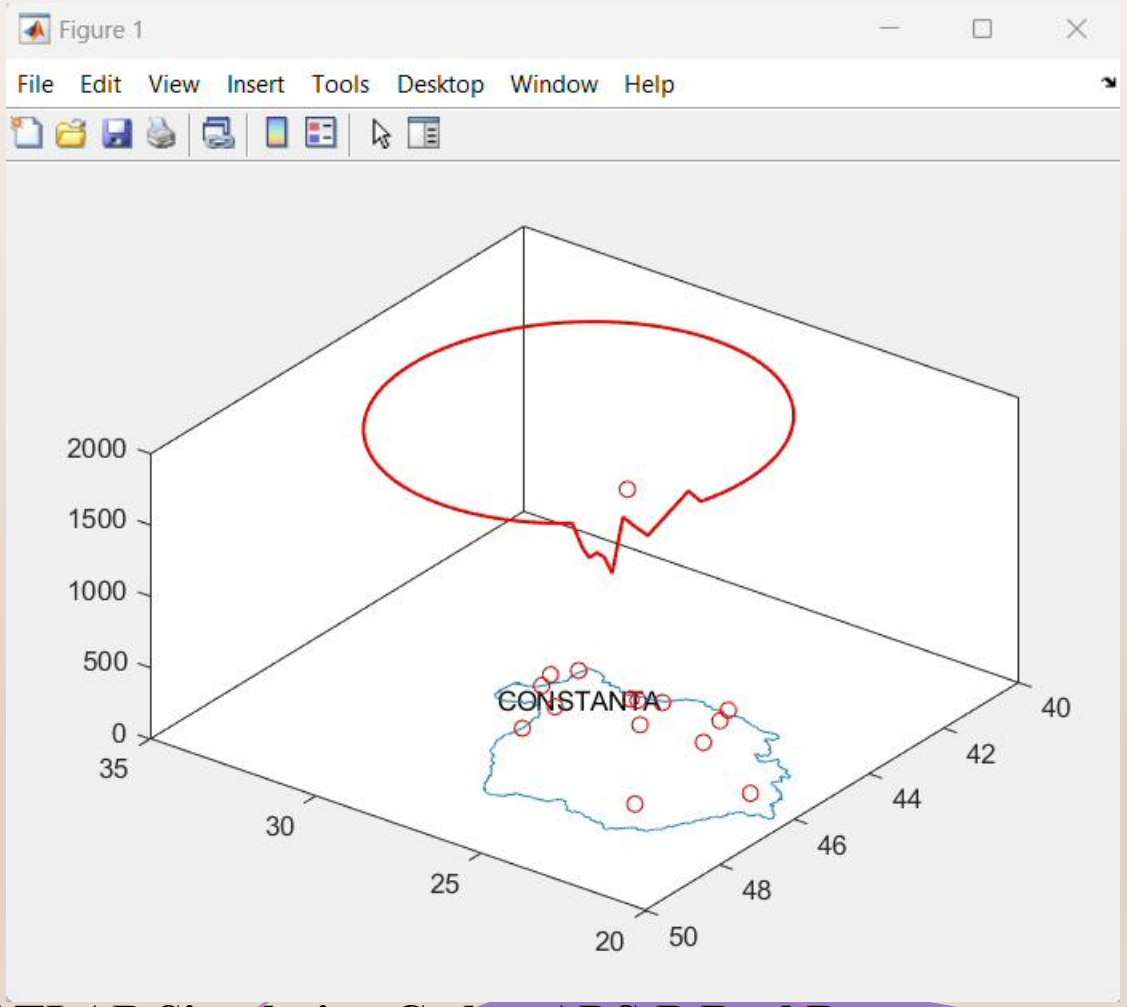



**MATLAB code is referenced in APPENDIX A-5: MATLAB Simulation Code – ADS-B Theoretical Range**

# ADS-B Real Range (1)

## Constanta ADS-B example

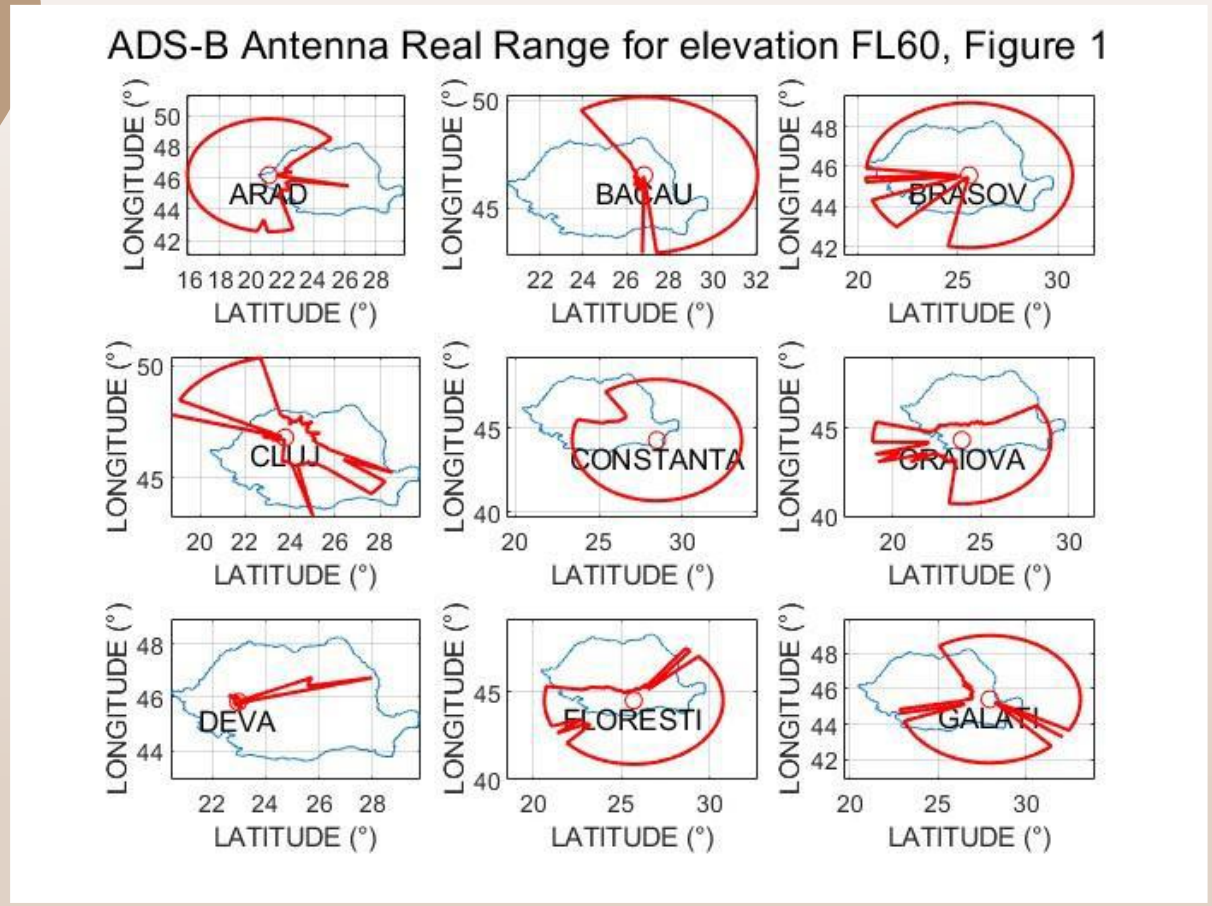


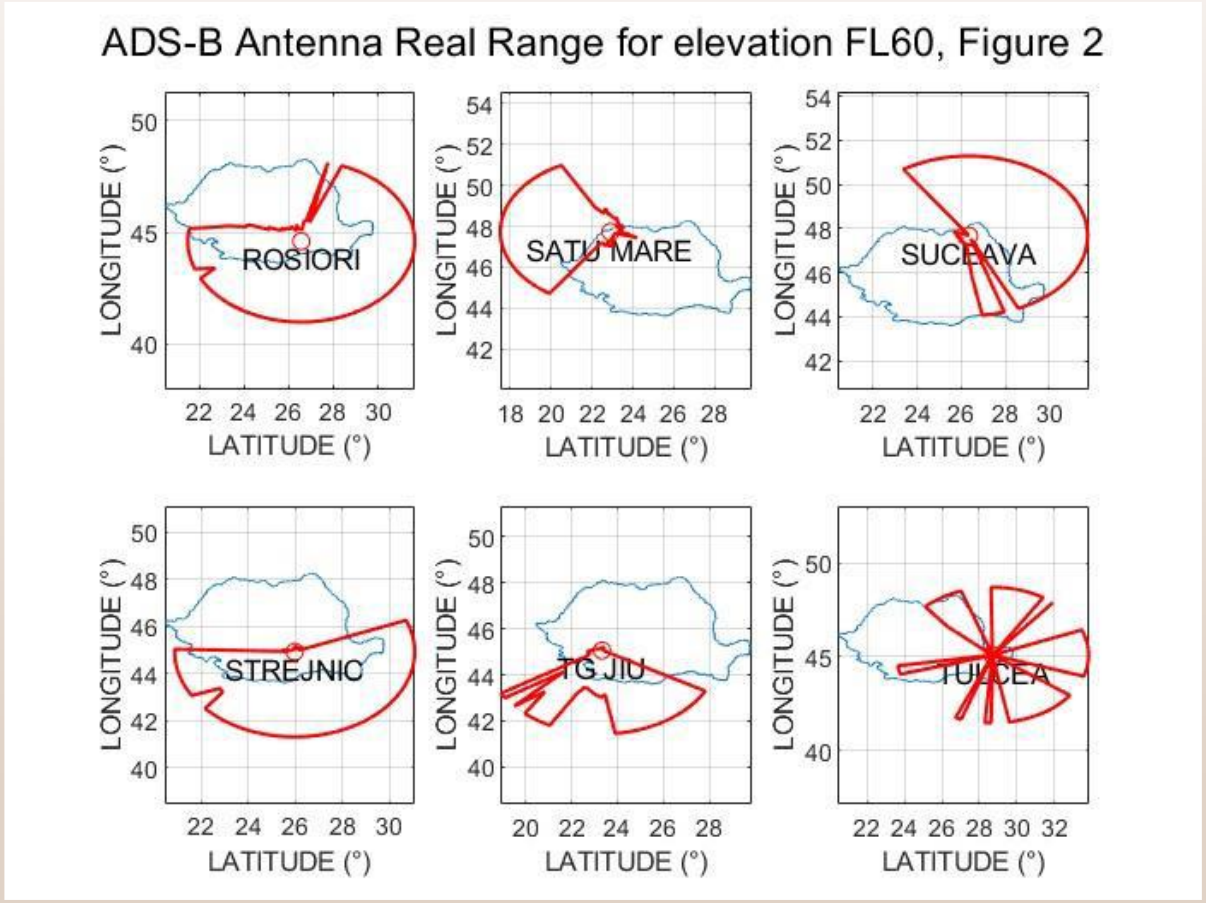




**MATLAB code is referenced in APPENDIX A-6: MATLAB Simulation Code – ADS-B Real Range**

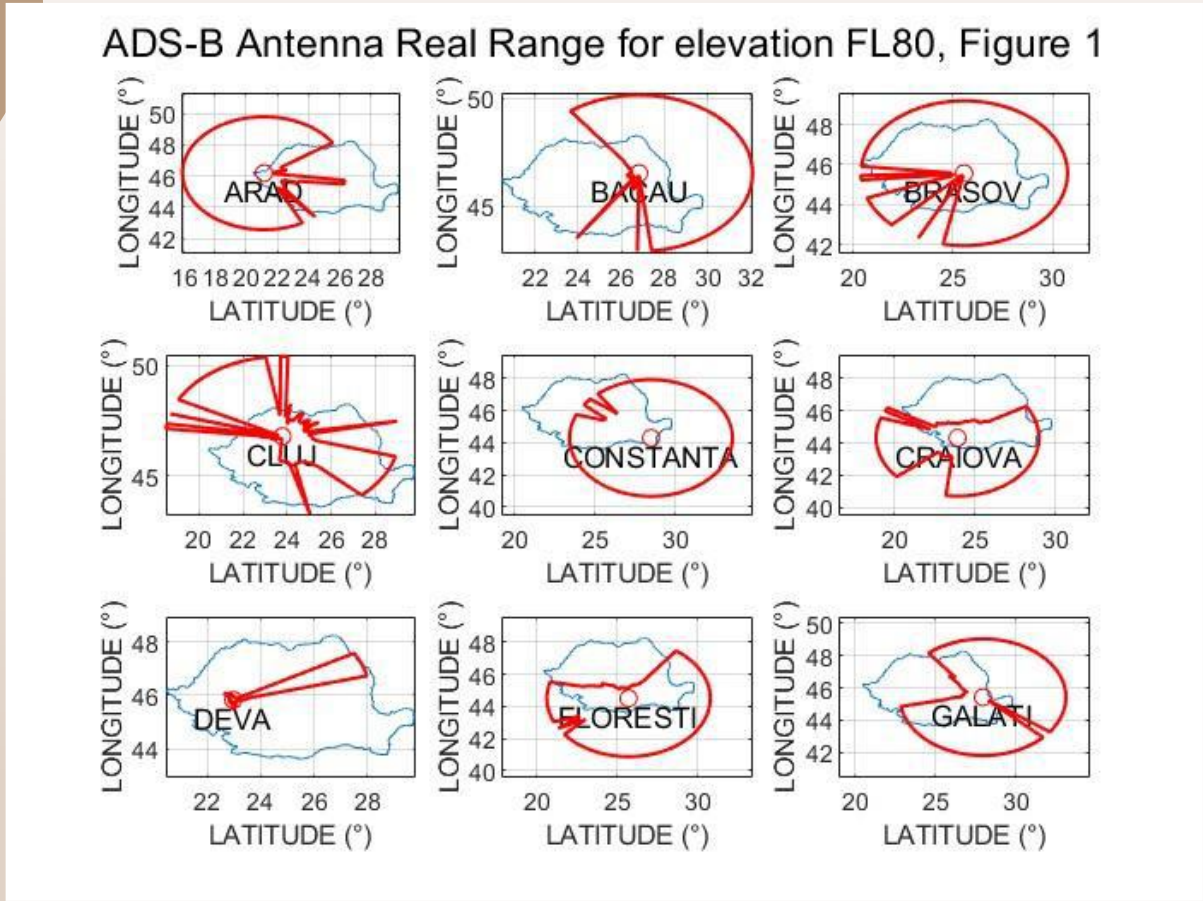
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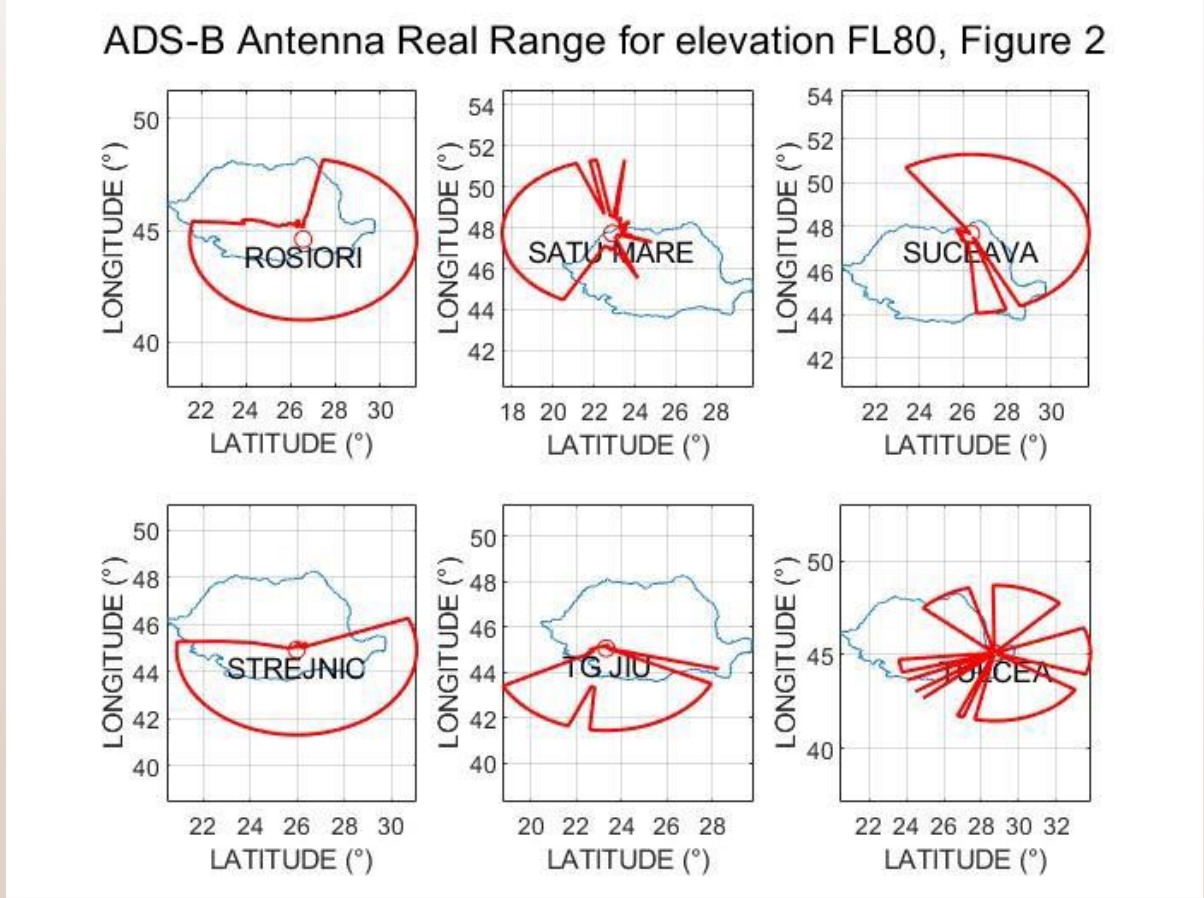
# ADS-B Real Range (2)



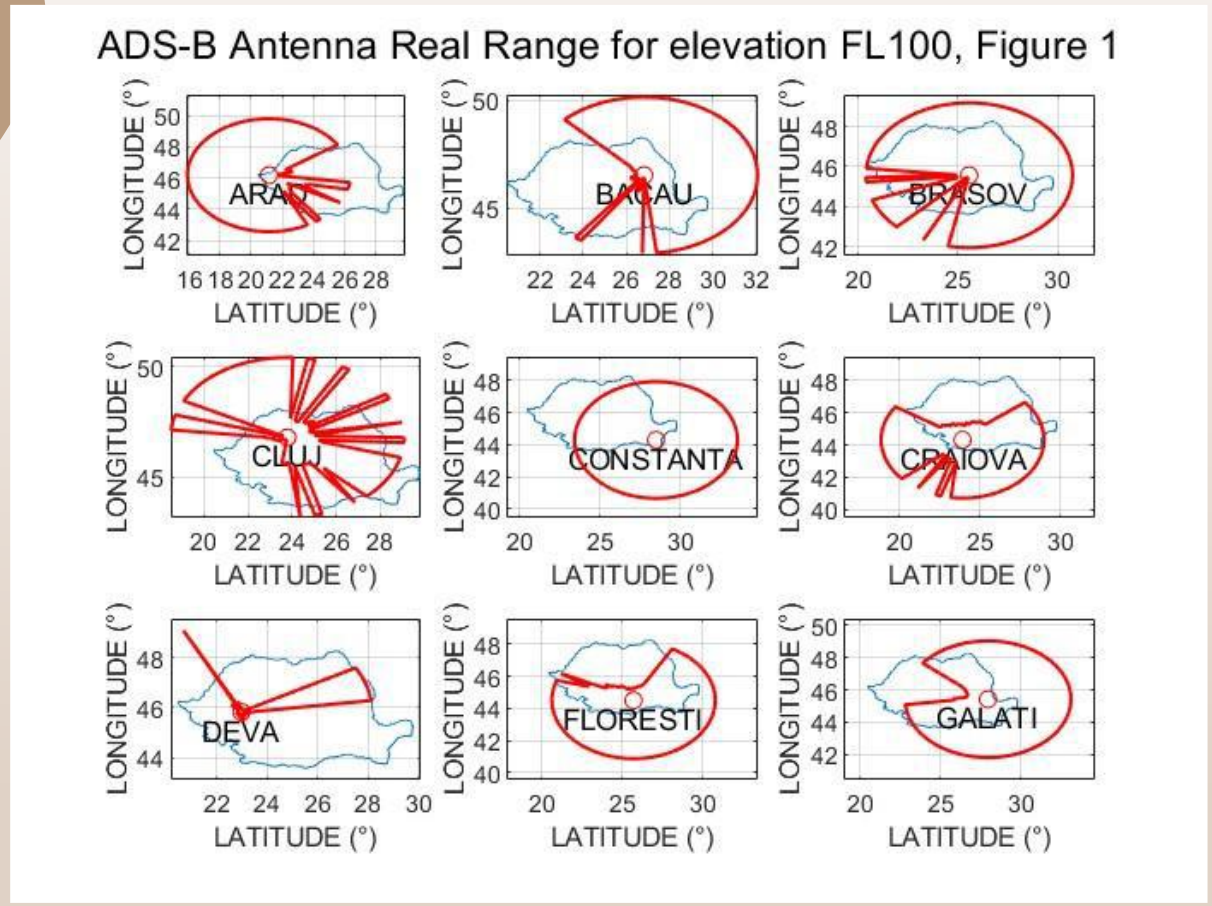



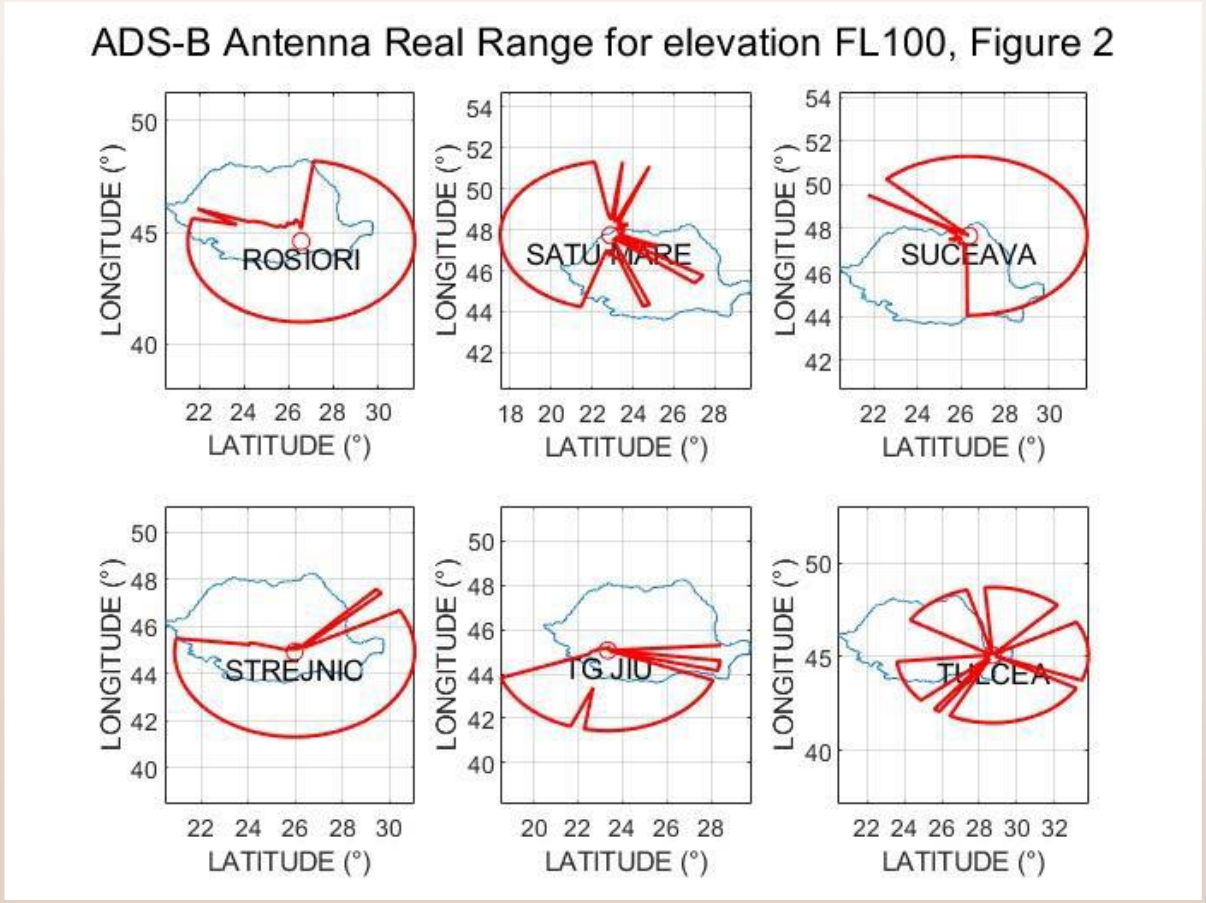
# ADS-B Real Range (3)



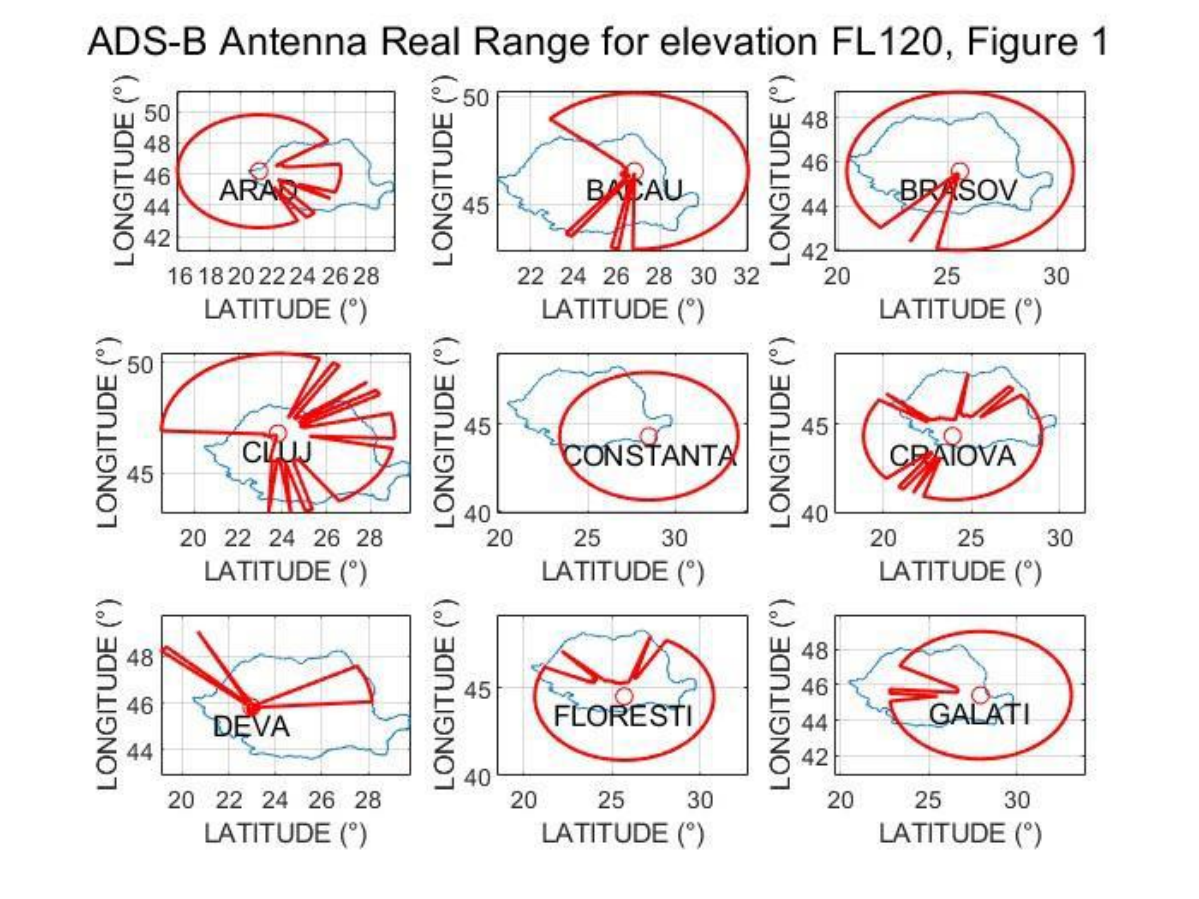



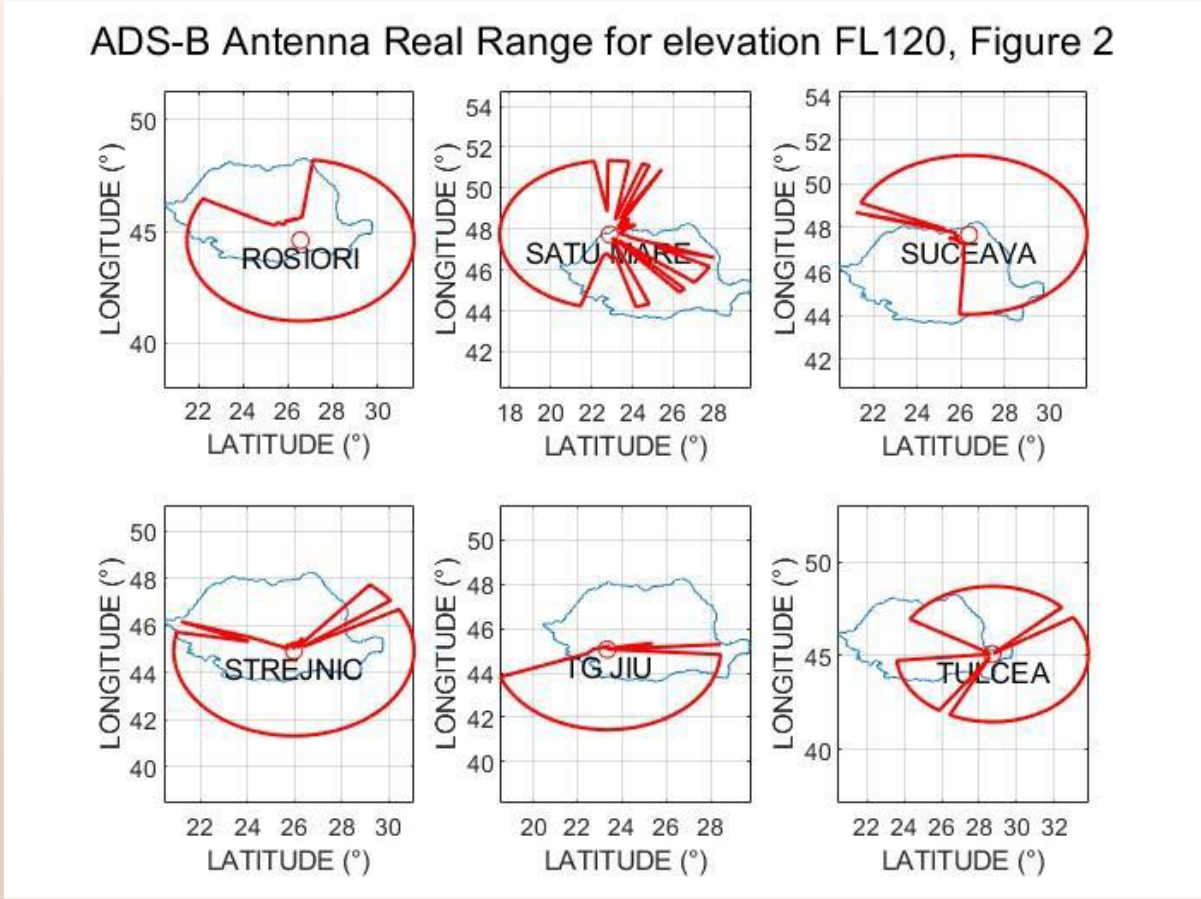
# ADS-B Real Range (4)



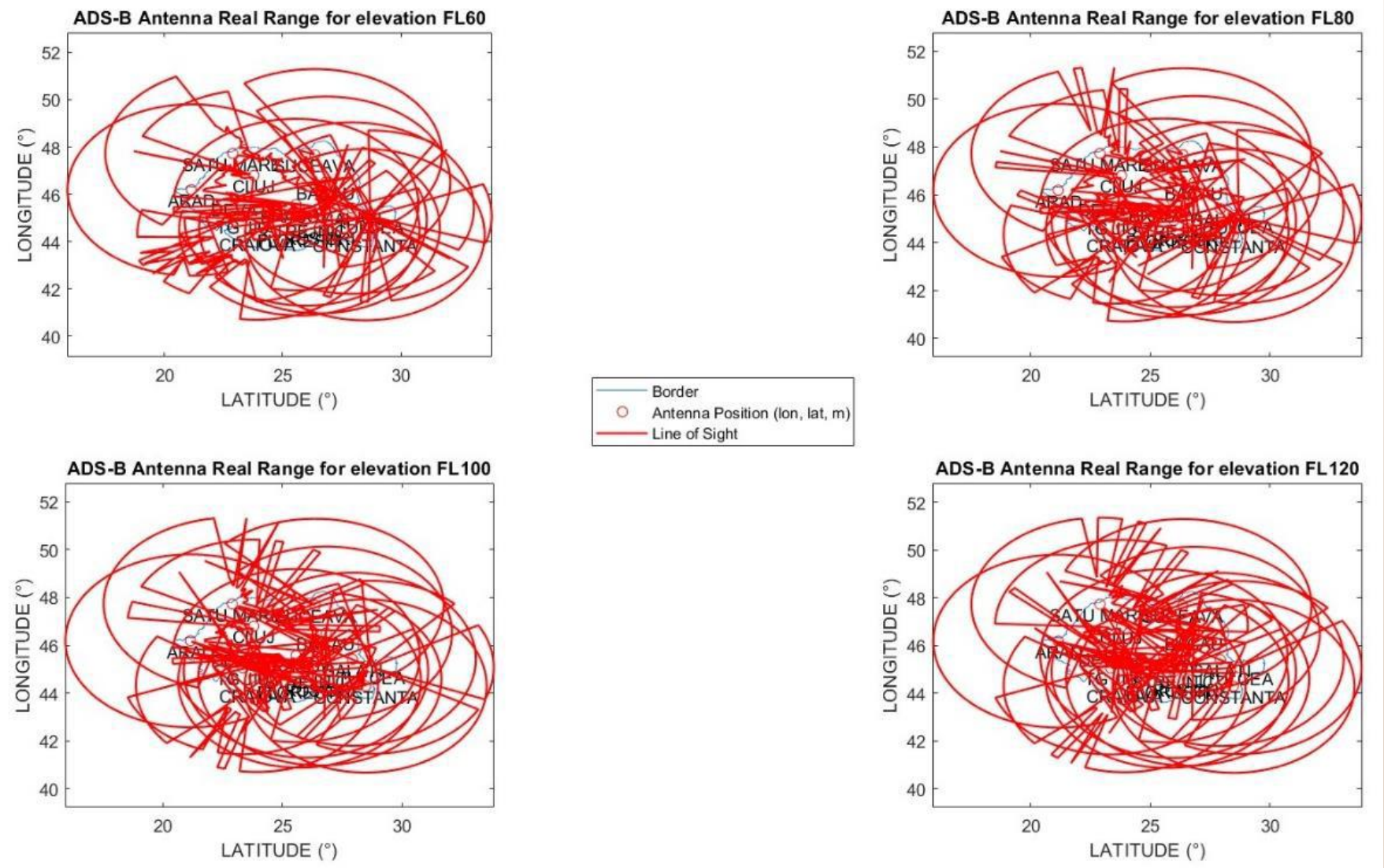



# ADS-B Real Range (5)



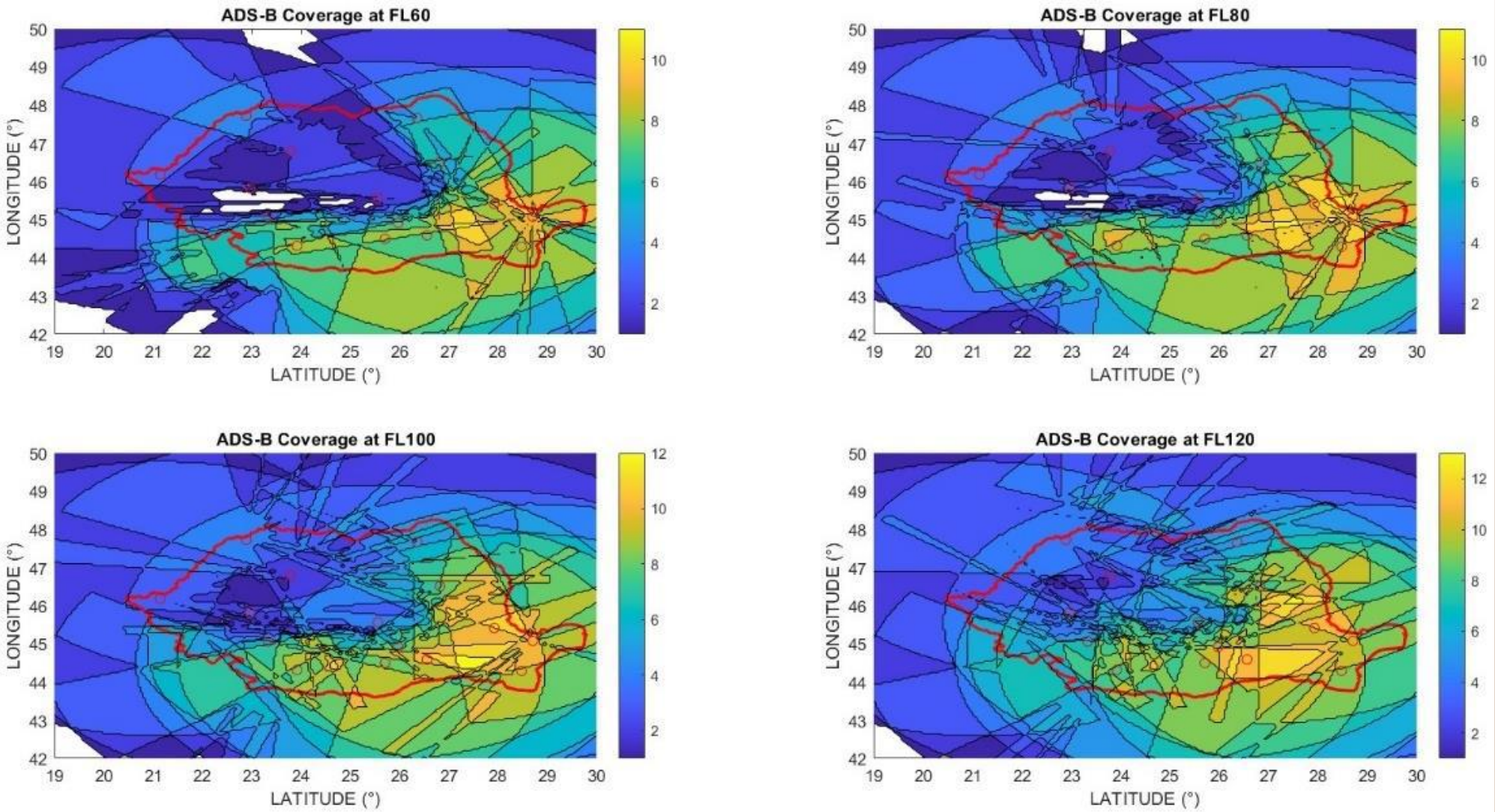
# ADS-B Real Range (6)




**MATLAB code is referenced in APPENDIX A-7: MATLAB Simulation Code – MATRIX**

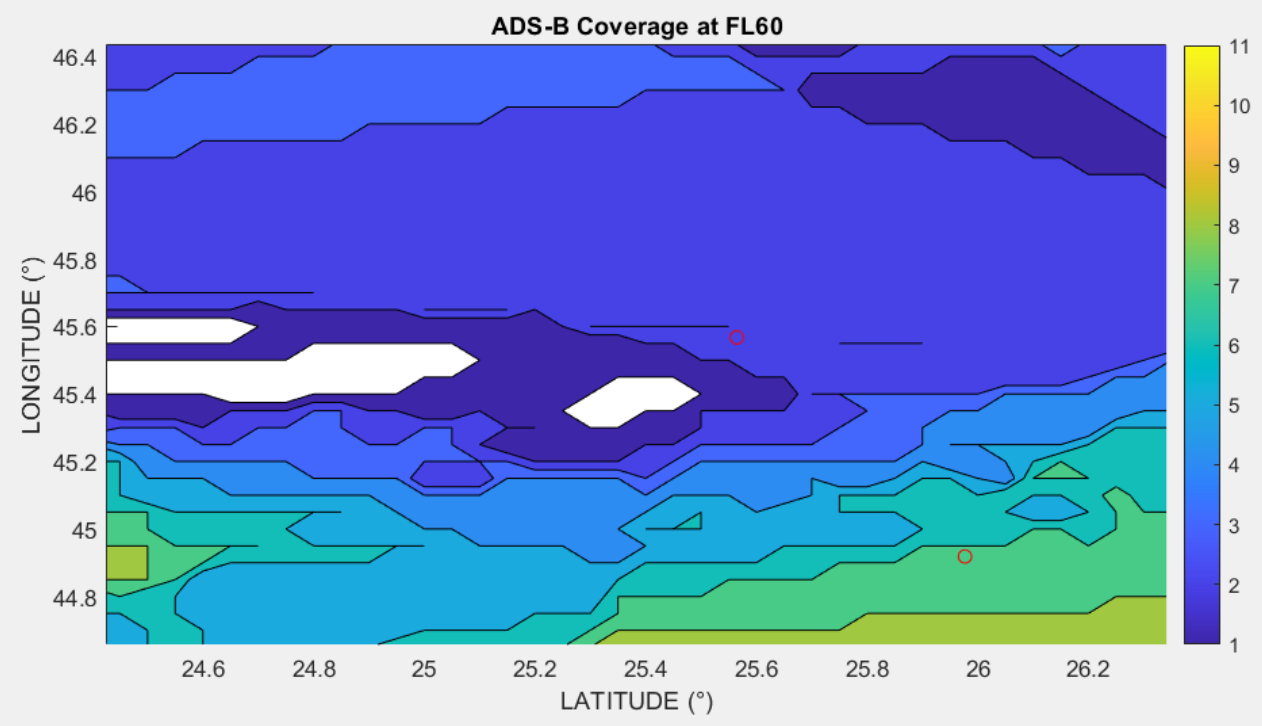
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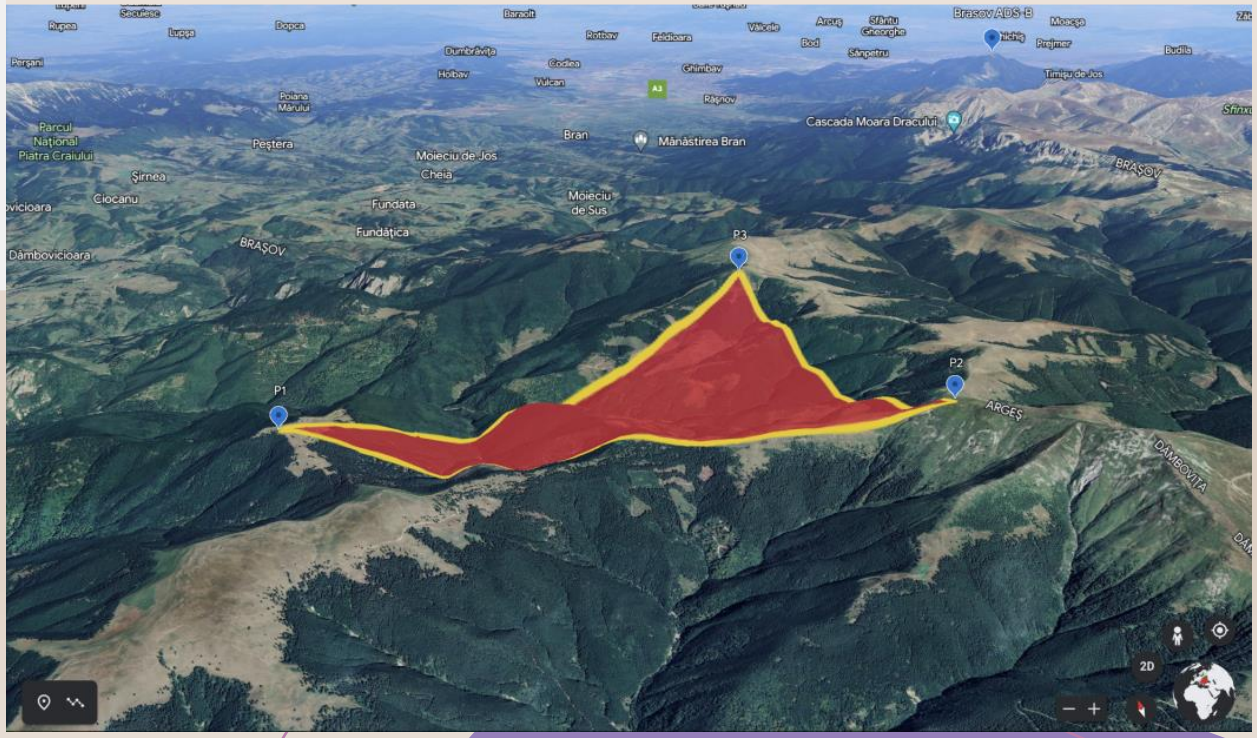
# Simplified view



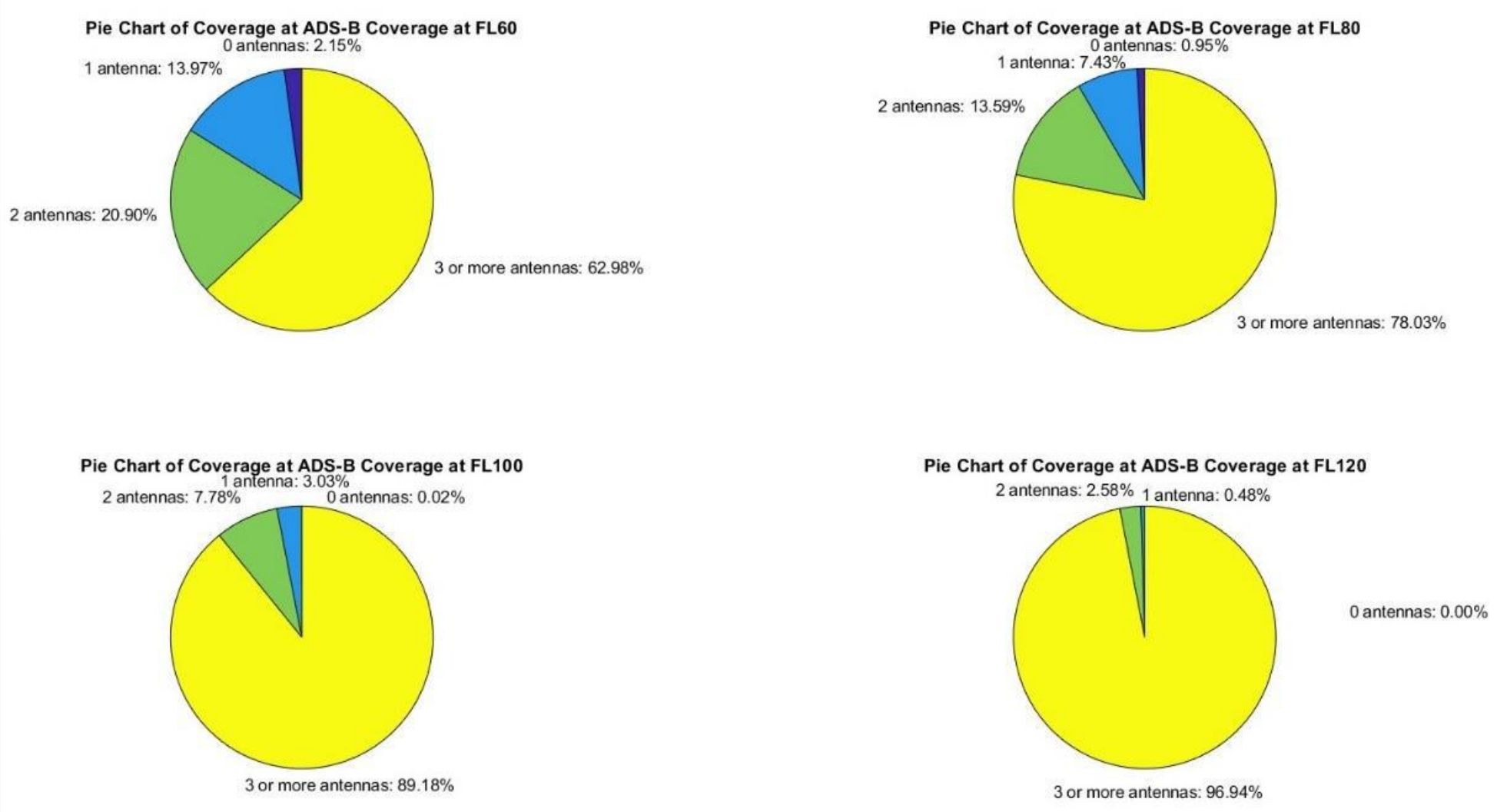

**MATLAB code is referenced in APPENDIX A-8: MATLAB Simulation Code – Coverage Map**

# Example of error study



# Result interpretation




**MATLAB code is referenced in APPENDIX A-9: MATLAB Simulation Code – Coverage Percentage**
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# Key findings

1. Significant higher ADS-B coverage in Muntenia, Dobrogea and Moldova
2. Carpathians mountain chain play a major role in ADS-B signal propagation in Romania
3. Impossible to have full coverage at relatively low flight levels
4. Mitigation of uncovered areas can be obtained with new infrastructure implementation
5. ADS-B must NOT replace RADAR systems but to complement it

# Future work

A. Infrastructure improvements for ADS-B signal in Romania

B. Research on Integrating MLAT with Other Surveillance Systems

C. Analyze ADS-B signals from neighbor countries

D. Relocate actual ADS-B antennas for improved coverage

# Bibliography


Bremberg, S., 2015. *Calibration of Multilateration Positioning Systems via Nonlinear Optimization.* 1st ed. Stockholm, Sweden: KTH Royal Institute of Technology.

Brij N. Agrawal, M. F. P., 2018. *Standard Handbook for Aerospace Engineers.* s.l.:McGraw-Hill Education.

Gaviria, I. A. M., 2013. *New strategies to improve multilateration systems in the air traffic control.* 1st ed. Valencia: Universitat Politecnica de Valencia.

International Air Transport Association (IATA), 2016. *Aviation Usages of Spectrum.* Montreal, Canada: s.n.

International Civil Aviation Organization (ICAO), 2021. *Mode S Downlink Aircraft Parameters Implementation and Operations Guidance Document.* s.l.:s.n.

International Telecommunication Union (ITU), 1996. *Radiowave Propagation Information for Predictions for Earth-to-Space Path Communications.* Geneva, Switzerland: s.n.

International Telecommunication Union (ITU), 2013. *Radiometeorology.* Geneva, Switzerland: s.n.

Internatonal Civil Aviation Organization - Asia and Pacific Office, 2007. *Multilateration (MLAT) - Concept of use.* s.l.:s.n.

Nagy, A., 2016. *Multilateration and Kalman Filtering Techniques for Stealth Intelligence Surveillance and Reconnaissance Using Multistatic Radar.* 1st ed. Ottawa, Ontario: Ottawa-Carleton Institute for Electrical Engineering.

National Aerospace Laboratory NLR, 2004. *Wide Area Multilateration Report on EATMP TRS 131/04,* s.l.: s.n.

Pleter, O. T., 2022. *Air Navigation Services Course.* Version 1.1 ed. Bucharest: Brainbond.

Pleter, O. T., 2022. *Radio Navigation Systems Course.* Version 5.2 ed. Bucharest, Romania: Braindbond.

Puričer, P., 2019. *ADS–B Signal Processing Improvement in Difficult Enviroment.* Prague: s.n.

Richards, M. A., 2014. *Fundamentals of Radar Signal Processing.* s.l.:McGraw-Hill Education.

Rudge, A. W. M. K. O. A. D. a. K., 1983. *The Handbook of Antenna Design.* London, U.K.: Peregrinus.

Sang-Hoon Yoo, J.-H. O. Y.-M. K. S.-H. K. T.-K. S., 2016. *Performance Analysis of MLAT System Receiver for Aircraft Flight Control System.* 5(1) ed. Korea: JPNT.

southwestantennas.com, 2020. *High Performance ADS-B Omni Antenna, 1090 MHz, 5.5 dBi, for ADS-B Receivers.* [Online] Available at: https://www.southwestantennas.com/omni-antennas/4-section-collinear/high-performance-ads-b-receiver-vertical-omni-antenna-1090-mhz-5-0

Stroman, S., 2021. *Automatic Dependent Surveillance Broadcast (ADS-B) Security Mitigation through Multilateration.* Florida, USA: UNF Graduate Theses and Dissertations.

Thales Group, 2013. *AX/BX680 Automatic Dependant Surveillance-Broadcast System.* [Online] Available at: https://www.thalesgroup.com/sites/default/files/database/d7/asset/document/ax680_bx680_datasheet.pdf

Thales, 2021. *Technical and Operational Considerations for ADS-B Implementation,* s.l.: s.n.

United States Geological Survey (USGS), 2023. *Earth Explorer.* [Online] Available at: https://earthexplorer.usgs.gov/

# Questions & Discussions